Logic, Language, and Philosophy

Proposition logic: semantics

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Martin Stokhof

Department of Philosophy, Tsinghua University, Beijing
ILLC/Department of Philosophy, University of Amsterdam

Today

Introduction

Truth tables

- Defining connectives
- Using truth tables

Valuations

Homework

Introduction

Validity: the semantic view

One way to define validity is as follows:

> *An argument is valid if, and only if, the truth of the premisses implies the truth of the conclusion*

In other words:

- If an argument is valid, then if the conclusion is false, at least one of the premisses is false.

This shows that validity is a matter of connections between the truth and falsity of statements

This is called a semantic definition of validity

Referentiality

Note:

- The definition of validity looks only at the truth values of premisses and conclusion
- Thus the content of premisses and conclusion is irrelevant

This is often called a ‘referential’ look at meaning

Let’s make this concrete for propositional languages

Proposition letters

First up: proposition letters:

- Proposition letters have truth values: 1, for ‘true’; 0, for ‘false’
- Truth values of proposition letters are independent

This means that proposition letters are semantic atoms

Complex formulas

Next: complex formulas

- The truth value of a complex formula is determined by
  a. the truth values of the proposition letters it contains
  b. the way it is constructed from proposition letters and connectives

Ad b.: each connective determines the truth value of the complex that it constructs from the truth values of the sub-formulas on which it operates

This implements the *principle of compositionality*:

> The meaning of a whole is a function of the meaning of its parts and the way in which these are put together

Given the meaning of a formula is its truth value, this means that the semantics of proposition logic is truth-functional

Two implementations

We present two ways of implementing a referential semantics for propositional languages:

- Using truth tables
- Using valuations

Truth tables

Two functions

A truth table is a way of defining, respectively investigating, systematic connections between the truth values of formulas.

Defining:

- Use a truth table to define the meaning of a connective

Investigating:

- Use truth tables to investigate properties of complex formulas
- Use truth tables to investigate validity of of argument schemes

Truth tables

Defining connectives

Negation

ϕ	$\neg\phi$
1	0
0	1

Table 1: Truth table for negation

Note the use of the meta-variable ϕ

Examples

- John is at home –John is not at home / John is out
- Everyone is welcome – Someone is not welcome
- It was the butler or the cook – It was not the butler and also not the cook
- It’s neither here nor there – It is somewhere

Conjunction

ϕ	ψ	$\phi \wedge \psi$
1	1	1
1	0	0
0	1	0
0	0	0

Table 2: Truth table for conjunction

Examples

- John stayed at home and so did Mary
- John and Mary stayed at home
- All three men left early – Man_1 and man_2 and man_3 left early

Note:

- John and Mary got married and had a baby.
- John and Mary had a baby and got married.

Orders matter in these examples, but the truth table does not reflect it

Conjunction in other guises

- John is at home but Peter is not
- Even though it was beautiful outdoors, John stayed indoors
- Although it was extremely cold, John did not stay indoors.

All examples express a conjunction but also say something additional about the relation between the conjuncts

Not conjunctions

- John and Peter are friends $\neq$ John is a friend and Peter is a friend
- Xiao Wang and Xiao Li are fun at parties $\neq$ Xiao Wang is fun at parties and Xiao Li is fun at parties

Disjunction

ϕ	ψ	$\phi \vee \psi$
1	1	1
1	0	1
0	1	1
0	0	0

Table 3: Truth table for disjunction

This is so-called *inclusive* disjunction: this disjunction also true of both disjuncts are true

Exclusive disjunction

ϕ	ψ	$\phi \bowtie \psi$
1	1	0
1	0	1
0	1	1
0	0	0

Table 4: Truth table for exclusive disjunction

This is so-called *exclusive* disjunction: this disjunction is false if both disjuncts are true

Inclusive and exclusive disjunctions

- Inclusive: One can learn English by watching films or by interacting with others
- Exclusive: Either John drives me to the airport or I get a taxi
- Exclusive: We are going to see a film tonight, unless we are going to the beach this afternoon.

Material implication

ϕ	ψ	$\phi \to \psi$
1	1	1
1	0	0
0	1	1
0	0	1

Table 5: Truth table for material implication

Implication and ‘If ..., then ...’

Example:

(A) If Bob drives too fast, he gets a fine

Consider the following cases:

1. Bob drives too fast and he gets a fine
2. Bob drives too fast and he does not get a fine

Clearly, sentence (A) is true in case 1. and false in case 2.

What about the other two cases, where the antecedent is false?

3. Bob did not drive too fast and he gets a fine
4. Bob did not drive too fast and he does not get a fine

Case 3. does not show that (A) is false. But (A) has to be either true of false, so we consider it true.

And the same holds for case 4.

Material implication ‘in the wild’

The implications we encounter in mathematics are material implications:

(B) If a number is larger than 5, then it is larger than 3

1. If 6> 5, then 6 > 3
2. If 4 > 5, then 4 > 3
3. If 2 > 5, then 2 > 3

The ‘false’ case cannot arise, which shows that the sentence is *always* true, which is as it should be.

Examples

- John is in a bad mood only if he has just gotten up
- In order for the party to function better, it is necessary that more contact be made with the electorate
- In order for the party to function better, it is sufficient that Smith be ousted

They can be rewritten as follows by using ‘ if ...then ...’:

- If John is in a bad mood, then he has just gotten up
- If the party functions better, then more contact is made with the electorate.
- If Smith is ousted, then the party functions better.

(Material) equivalence

ϕ	ψ	$\phi \leftrightarrow \psi$
1	1	1
1	0	0
0	1	0
0	0	1

Table 6: Truth table for material equivalence

Truth tables

Using truth tables

Truth tables for complex formulas

Truth tables can also be used to specify the truth conditions of complex formulas ϕ, i.e., the conditions under which ϕ is true

What defines those conditions?

- The distributions of truth values over the proposition letters in ϕ
- The composition of ϕ using connectives, as laid down in ϕ's construction tree

Next week:

- In terms of truth conditions we can then investigate properties of formulas and validity of arguments schemes

Composite truth tables

p	q	$\neg p$	$\neg q$	$\neg p \to \neg q$	$\neg(\neg p \to \neg q)$
1	1	0	0	1	0
1	0	0	1	1	0
0	1	1	0	0	1
0	0	1	1	1	0

Table 7: Composite truth table for $\neg(\neg p \to \neg q)$

Composite truth tables (2)

p	q	r	$p \vee q$	$\neg r$	$(p \vee q) \wedge \neg r$
1	1	1	1	0	0
1	1	0	1	1	1
1	0	1	1	0	0
1	0	0	1	1	1
0	1	1	1	0	0
0	1	0	1	1	1
0	0	1	0	0	0
0	0	0	0	1	0

Table 8: Composite truth table for $(p \vee q) \wedge \neg r$

Procedure

1. Determine which n distinct propositions letters $p_1 \dots p_n$ occur in ϕ
2. Draw a table with 2^n rows and fill each row with a unique combination of truth values assigned to $p_1 \dots p_n$
3. Follow the construction tree of ϕ and add a column for each sub-formula of ϕ, *including ϕ itself*
4. Calculate for each column the truth values for each row

Valuations

Valuations

An alternative definition of the semantics of propositional languages uses valuations:

Definition (Semantics)

A valuation V is a function from FORM_L to $\{0, 1\}$ that meets the following requirements:

1. $V(p_i) \in \{0, 1\}$, for all $p_i \in \text{VOC}_L$
2. $V(\neg\phi) = 1$ iff $V(\phi) = 0$
3. $V(\phi \wedge \psi) = 1$ iff $V(\phi) = 1$ and $V(\psi) = 1$
4. $V(\phi \vee \psi) = 1$ iff $V(\phi) = 1$ or $V(\psi) = 1$
5. $V(\phi \to \psi) = 1$ iff $V(\phi) = 0$ or $V(\psi) = 1$
6. $V(\phi \leftrightarrow \psi) = 1$ iff $V(\phi) = V(\psi)$

Note: clauses for connectives match their truth table definitions

From semantics to logic

We now have a semantics, but we do not yet have a logic, i.e., a formal definition of validity of argument schems

That will the topic of the next logic session

First: discussion session about texts from Suber and Sandford

Topic: the semantics and the logic of conditionals

Homework

Homework

Homework for next session on Saturday October 11

Read:

- Chapters 5 and 6
- Texts by Peter Suber and David Sandford

Homework for next logic session on Monday October 13:

From chapter 5, make the following exercises:

- Exercises 1, 2, and 3

Enjoy the holiday!